Week 5: Basic electromagnetism in vacuum and solids

Physics classes, M1 introductory module
Master in Molecular Nanoscience and Nanotechnology
ICMol Seminar room 0.10.6

Topics:

Electric and magnetic forces. Maxwell equations and electromagnetic waves.

Timetable:

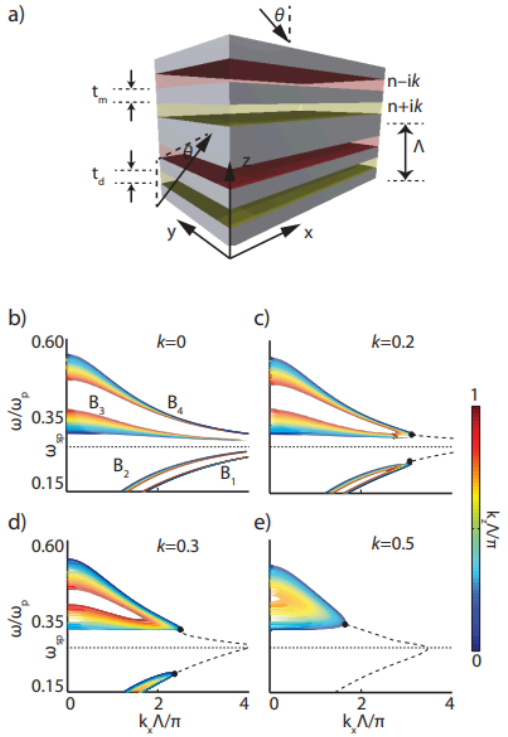
6th of November

10.00–10.15	Brief introduction to today's class: why care about electromagnetism? **Efrén Navarro**
10.15–10.35	Maxwell's displacement current and the magnetic field between capacitor electrodes T. Hyodo *European Journal of Physics* **43**, 065202 (2022). **Lucía Navas** Full paper: page 6
10.35–11.30	Lecture, discussion and revision of concepts.
11.30–11.45	*Break*
11.45–12.05	Parity-time-symmetric plasmonic metamaterials H. Alaeian et al. *Physical Review A* **89**(3), 033829 (2014). **Ángela Cuenca** Full paper: page 20
12.05–13.15	Lecture, discussion and revision of concepts.

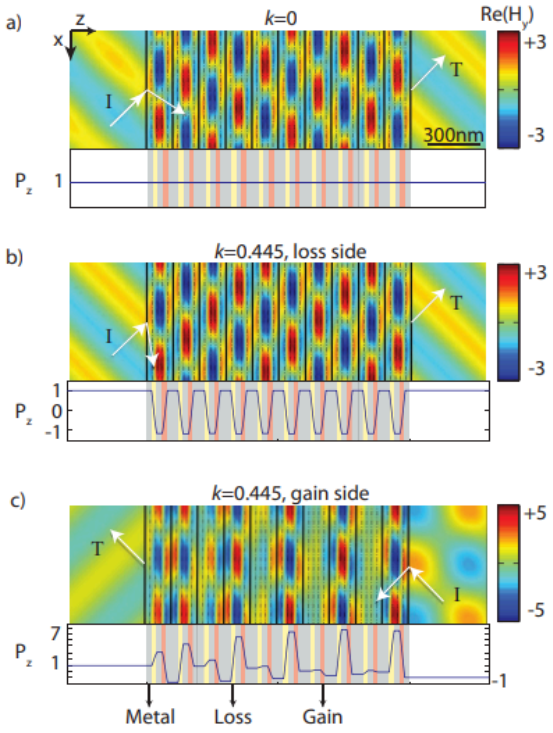
Journal club handouts

P T-Symmetric Plasmonic Metamaterials

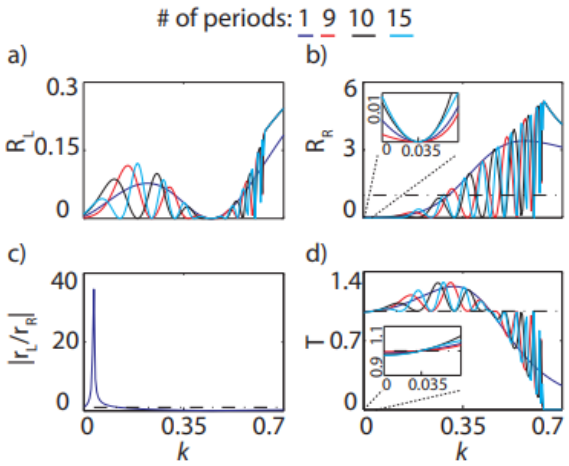
1.



2.



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Full papers

Topic: Electric and magnetic forces

Time: 10.15–10.35

Maxwell's displacement current and the magnetic field between capacitor electrodes

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Lucía Navas

European Journal of Physics

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Maxwell's displacement current and the magnetic field between capacitor electrodes

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Abstract

A long-standing controversy concerning the causes of the magnetic field in and around a parallel-plate capacitor is examined. Three possible sources of contention are noted and detailed. The first is the ambiguous initial impression given by the calculation of the magnetic field using the integral form of the Ampere–Maxwell law which incorporates the displacement current density. The second is misinterpretation of this law as a cause–effect formula. The third is insufficient recognition of the fact that the electric field in Maxwell's equations represents the sum of the well-distinguished irrotational and divergence-free fields, which are independently responsible for conservation of charge and the existence of the electromagnetic waves, respectively.

Keywords: displacement current, magnetic field, Maxewll's equations, Ampere–Maxwell law, Biot–Savart law, electric field, parallel-plate capacitor

(Some figures may appear in colour only in the online journal)

1. Introduction

The displacement current density introduced by Maxwell in his theory of electromagnetism has long been a topic of debate. (Although the concept of the electric displacement $\boldsymbol{D}(\boldsymbol{r}, t)$ already carries a notion of *surface density*, here for clarity we call $\partial \boldsymbol{D}(\boldsymbol{r}, t)/\partial t$ the displacement current density and its surface integral the displacement current.) A typical case of contention

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is whether the magnetic field in and around the space between the electrodes of a parallel-plate capacitor is created by the displacement current density in the space. History of the controversy was summarized by Roche [1], with arguments that followed [2–4] showing the subtlety of the issue. Notwithstanding that Maxwell himself thought that the displacement current gave rise to the magnetic field (just as the conduction current did) [5], it was later verified not to be the case for the part of the displacement current density that is proportional to the time derivative of the conservative (irrotational or longitudinal) electric field related to the scalar potential [6–10]. It would be worth noting that the points of contention in the apparently complicated arguments [1–4] between Roche and Jackson were substantially on the details, subject to agreement on the main point that $\partial \boldsymbol{D}/\partial t$ between capacitor electrodes is not the true source of the magnetic field.

However this characteristic of the displacement current density has not been widely recognized yet compared with the following two: (i) that it makes Maxwell's equations compatible with the conservation of charge; and (ii) that it plays a crucial role in deriving the differential equation for the electromagnetic wave. In the present article I revisit this topic and explore the issue in some detail to clarify perception and to prevent confusion. It is hoped that the discussion is accessible to undergraduate students with the aim of engendering a clear understanding of this point from when first encountering the notion of displacement current density. Thus I avoid discussions requiring the use of the electromagnetic (scalar and vector) potentials.

The following section introduces the displacement current density and the application of the resultant Ampere–Maxwell law to calculate magnetic field (commonly found in textbooks and lectures). In section 3, use of the Biot–Savart law and Ampere–Maxwell law for the calculation of the magnetic field is discussed. The experiments claiming to have observed the displacement current are briefly reviewed in section 4. In section 5, Maxwell's equations are carefully deciphered from the reference point that the electric field in the equations is the sum of the conservative field and the electromagnetic induction field, with the aim of clearing up misconceptions about the role of displacement current density. A summary is given in section 6.

2. Calculation of the magnetic field using the integral form of the Ampere–Maxwell law

Before we proceed, let us summarize Maxwell's equations. The electric field $\boldsymbol{E}(\boldsymbol{r}, t)$ and the magnetic field $\boldsymbol{B}(\boldsymbol{r}, t)$ at a position $\boldsymbol{r}$ and time t in an inertial frame of reference is defined by the Lorentz force $\boldsymbol{F}$ on a charge q moving at a velocity $\boldsymbol{v}$

$$\boldsymbol{F} = q(\boldsymbol{E}(\boldsymbol{r}, t) + \boldsymbol{v} \times \boldsymbol{B}(\boldsymbol{r}, t)). \tag{1}$$

This means that $\boldsymbol{E}(\boldsymbol{r}, t)$ is defined from the force on a charge q regardless of its velocity and $\boldsymbol{B}(\boldsymbol{r}, t)$ is defined from the force on a moving charge only.

The other relevant fields, the electric displacement (electric flux density) $\boldsymbol{D}(\boldsymbol{r}, t)$ and the magnetic field intensity $\boldsymbol{H}(\boldsymbol{r}, t)$, are related with the sources of the fields. The relationships of these fields in free space are

$$\boldsymbol{D}(\boldsymbol{r}, t) = \varepsilon_0 \boldsymbol{E}(\boldsymbol{r}, t) \tag{2}$$

$$\boldsymbol{H}(\boldsymbol{r}, t) = {\mu_0}^{-1} \boldsymbol{B}(\boldsymbol{r}, t) \tag{3}$$

where ε_0 is the permittivity of free space (electric constant) and μ_0 is the permeability of free space (magnetic constant).

By using these four fields together with charge density $\rho(\boldsymbol{r},t)$ and current density $\boldsymbol{j}(\boldsymbol{r},t)$, Maxwell's equations, in both integral and differential forms, are written as follows:

$$\oint_{\mathrm{S}} \boldsymbol{D}\cdot \mathrm{d}\boldsymbol{S} = \int_{\mathrm{V}} \rho \,\mathrm{d}V, \qquad \nabla\cdot\boldsymbol{D} = \rho \tag{4}$$

$$\oint_{\mathrm{C}} \boldsymbol{E}\cdot \mathrm{d}\boldsymbol{r} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathrm{S}} \boldsymbol{B}\cdot \mathrm{d}\boldsymbol{S}, \qquad \nabla\times\boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{5}$$

$$\oint_{\mathrm{S}} \boldsymbol{B}\cdot \mathrm{d}\boldsymbol{S} = 0, \qquad \nabla\cdot\boldsymbol{B} = 0 \tag{6}$$

$$\oint_{\mathrm{C}} \boldsymbol{H}\cdot \mathrm{d}\boldsymbol{r} = \int_{\mathrm{S}} \left(\boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}\right)\cdot \mathrm{d}\boldsymbol{S}, \quad \nabla\times\boldsymbol{H} = \boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t} \tag{7}$$

where, S in (4) and (6) indicates the closed surface enclosing an arbitrary volume V, and S in (5) and (7) indicates arbitrary open surface bounded by an arbitrary closed path C. In (5) and (7) $\mathrm{d}\boldsymbol{S}(=\boldsymbol{n}\,\mathrm{d}S)$ is the surface element having the normal $\boldsymbol{n}$ directed in accordance with the right-hand rule with the direction of $\mathrm{d}\boldsymbol{r}$ in the circulation integral along C from the left-hand side of the equations. Note, I have dropped the notation $(\boldsymbol{r},t)$ in the equations that hold at any position and time in a frame of reference.

It is well known that Maxwell added the displacement current density term $\partial\boldsymbol{D}/\partial t$ to Ampere's law,

$$\oint_{\mathrm{C}} \boldsymbol{H}\cdot \mathrm{d}\boldsymbol{r} = \int_{\mathrm{S}} \boldsymbol{j}\cdot \mathrm{d}\boldsymbol{S}, \qquad \nabla\times\boldsymbol{H} = \boldsymbol{j} \tag{8}$$

obtaining (7), now called the Ampere–Maxwell law, to make it work for the case of an open circuit containing a capacitor as shown in figure 1. It is assumed that the resistance of the leads and the electrodes is negligible.

If the circuit is a long straight line without the capacitor and with a current I flowing, one may apply (8) to find the magnetic field at point P_1, distance R away from the current. Applying the integral form of the law to a circle C_1 centered on the current and through the point P_1 together with flat surface S_1 perpendicular to the current and bordered by C_1, one gets

$$\oint_{\mathrm{C}_1} \boldsymbol{H}\cdot \mathrm{d}\boldsymbol{r} = \int_{\mathrm{S}_1} \boldsymbol{j}\cdot \mathrm{d}\boldsymbol{S} = I. \tag{9}$$

Noting the axial symmetry, one gets the magnitude of the magnetic field intensity at P_1 as

$$H_1 = \frac{I}{2\pi R}. \tag{10}$$

In the case shown in figure 1 where we have the capacitor, if P_1 is located far away from the capacitor, $\boldsymbol{H}$ has the azimuthal component alone and so the field is calculated in the same way to obtain (10). Use of S_2 instead of S_1 gives the same result. However, if one chooses the surface S_3,

$$\oint_{\mathrm{C}_1} \boldsymbol{H}\cdot \mathrm{d}\boldsymbol{r} = \int_{\mathrm{S}_3} \boldsymbol{j}\cdot \mathrm{d}\boldsymbol{S} = 0, \tag{11}$$

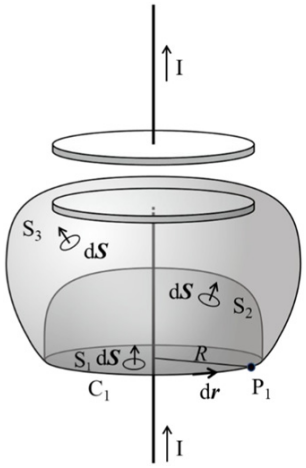


Figure 1. A circular parallel-plate capacitor being charged by the current I in long straight wires. A circle C_1 of radius R and surfaces S_1–S_3 bordered by C_1 are used to calculate the magnetic field at point P_1 on C_1. The surface element vectors $d\boldsymbol{S}$ for the surfaces S_1–S_3 are also shown. Their directions follow the right-hand rule with the direction of the line element $d\boldsymbol{r}$ for the circle C_1.

since no conduction current pierces S_3. This shows that Ampere's law cannot be used in the case of an open circuit including a capacitor. Maxwell's introduction of the displacement current density term fixed this difficulty. Applying now the integral form of (7) to S_3, one gets

$$\oint_{C_1} \boldsymbol{H} \cdot d\boldsymbol{r} = \int_{S_3} \frac{\partial \boldsymbol{D}}{\partial t} \cdot d\boldsymbol{S}. \tag{12}$$

Here we remember the law of charge conservation (continuity equation),

$$I + \frac{dQ}{dt} = \oint_S \boldsymbol{j} \cdot d\boldsymbol{S} + \int_V \frac{\partial \rho}{\partial t} dV = 0, \qquad \nabla \cdot \boldsymbol{j} + \frac{\partial \rho}{\partial t} = 0 \tag{13}$$

where S is the closed surface enclosing an arbitrary volume V, Q is the total charge in the volume V and I is the total flow of current out of the volume V. Then, with Gauss's law (4), the right-hand side of (12) is

$$\int_{S_3} \frac{\partial \boldsymbol{D}}{\partial t} \cdot d\boldsymbol{S} = \frac{d}{dt} \int_{S_3} \boldsymbol{D} \cdot d\boldsymbol{S} = \frac{d}{dt} \oint_{\overline{S}_1 + S_3} \boldsymbol{D} \cdot d\boldsymbol{S} = \frac{dQ}{dt} = I \tag{14}$$

where $\overline{S}_1 + S_3$ indicates the closed surface comprising of $\overline{S}_1$ and S_3, $\overline{S}_1$ being the same surface as S_1 but with the normal directed in the direction opposite to the case in (9) indicated by $d\boldsymbol{S}$ for S_1 in figure 1, and Q is the total electric charge inside the volume bounded by $\overline{S}_1 + S_3$, i.e., that on the bottom electrode. The term after the second equal sign in equation (14) holds since we assume that S_1 is far enough away from the capacitor that $\boldsymbol{D} = 0$ on it and the value of the surface integral does not change when $\overline{S}_1$ is added to the integral area. Equations (12) and (14) show that the choice of S_3 gives the same result as (9) using S_1 and S_2 owing to the displacement current density term.

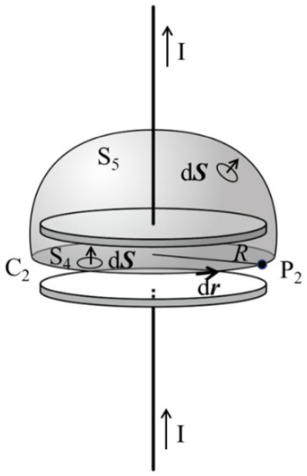


Figure 2. The same setup as figure 1. Point P_2 is on a circle C_2 of radius R, centered on the middle of the virtual straight line connecting the linear currents, and parallel to the capacitor electrodes.

Now let us consider the application of the law to calculate the magnetic field intensity H_2 at point P_2 in figure 2 using the surface S_4,

$$\oint_{C_2} \boldsymbol{H} \cdot \mathrm{d}\boldsymbol{r} = \int_{S_4} \frac{\partial \boldsymbol{D}}{\partial t} \cdot \mathrm{d}\boldsymbol{S} \approx I \tag{15}$$

leading to

$$H_2 \approx \frac{I}{2\pi R}. \tag{16}$$

The approximation (actually $H_2 \leqslant H_1$) is due to a fraction of the fringing electric field not piercing S_4 [7, 10] and approaches equality as R increases or as the distance between the electrodes decreases. This calculation may give the confusing first impression that the magnetic field there is created by $\partial \boldsymbol{D}/\partial t$ in the space between the electrodes. Although it is certainly plausible to calculate the magnetic field by using (15), it does not mean that the impression is correct, for the same field can be calculated by using the surface S_5 to which the main contribution comes from the conduction current I (rest is the partially cancelling contribution from the fringing $\partial \boldsymbol{D}/\partial t$). Mathematical plausibility of the calculation using the surface S_4 and S_5 is warranted by the close correlation between the two fields, $\boldsymbol{B}$ and $\partial \boldsymbol{D}/\partial t$, both of which result from the same conduction currents flowing in the leads and the electrodes (see section 3). Thus, both in figures 1 and 2, the *choice* of the surfaces in the calculation is irrelevant to whatever is the source of the field at point P_1 or P_2. Drawing students' attention to these points explicitly when first introducing the use of the integral form of the Ampere–Maxwell law will considerably reduce their misunderstanding.

3. Biot–Savart law and Ampere–Maxwell law

Another law that one may use to calculate magnetic field is the Biot–Savart law

$$\boldsymbol{H}(\boldsymbol{r}) = \int \frac{\boldsymbol{j}(\boldsymbol{r}') \times (\boldsymbol{r} - \boldsymbol{r}')}{4\pi|\boldsymbol{r} - \boldsymbol{r}'|^3} \mathrm{d}V'. \tag{17}$$

Expressions for surface current density $\boldsymbol{J}_\mathrm{S}(\boldsymbol{r}')$ flowing over a surface S or a current I flowing in a thin wire may be obtained by replacing $\boldsymbol{j}(\boldsymbol{r}')\mathrm{d}V'$ with $\boldsymbol{J}_\mathrm{S}(\boldsymbol{r}')\mathrm{d}S'$ or $I\,\mathrm{d}\boldsymbol{r}'$ as

$$\boldsymbol{H}(\boldsymbol{r}) = \int_\mathrm{S} \frac{\boldsymbol{J}_\mathrm{S}(\boldsymbol{r}') \times (\boldsymbol{r} - \boldsymbol{r}')}{4\pi|\boldsymbol{r} - \boldsymbol{r}'|^3} dS', \qquad \boldsymbol{H}(\boldsymbol{r}) = \oint_\mathrm{C} \frac{I\,\mathrm{d}\boldsymbol{r}' \times (\boldsymbol{r} - \boldsymbol{r}')}{4\pi|\boldsymbol{r} - \boldsymbol{r}'|^3} \tag{18}$$

respectively. This law describes how current density or a current element at $\boldsymbol{r}'$ contributes to the magnetic field at a point $\boldsymbol{r}$. (In Maxwell's equations as well, the expressions involving $\boldsymbol{j}$ and ρ are only valid when they are finite. In the case of surface density and line intensity where $\boldsymbol{j}$ and ρ are ∞, only the integral forms hold, applying the conversions shown above. This usual convention is used throughout the present paper.) This law is also applicable to an open circuit with a capacitor by adding up the contributions from all the conduction currents including, in the present case, the current spreading/centering radially over the electrode plates to make the charge on the plates increase/decrease [1, 2, 7, 9–12] as shown in figure 3. It is not only applicable to a static case but also approximately to a quasi-static case where the finiteness of the speed of light may be ignored [13, 14]. Furthermore, generalization of the Biot–Savart law to the time-dependent situation

$$\boldsymbol{B}(\boldsymbol{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\boldsymbol{j}(\boldsymbol{r}', t_\mathrm{r})}{R^2} + \frac{\partial \boldsymbol{j}(\boldsymbol{r}', t_\mathrm{r})/\partial t_\mathrm{r}}{cR} \right] \times \frac{\boldsymbol{R}}{R} \mathrm{d}V' \tag{19}$$

has been developed by Jefimenko [15–20], where $\boldsymbol{R} = \boldsymbol{r} - \boldsymbol{r}'$, c is the speed of light and

$$t_\mathrm{r} = t - \frac{|\boldsymbol{r} - \boldsymbol{r}'|}{c} = t - \frac{R}{c} \tag{20}$$

is the retarded time (former time). It reduces to the original Biot–Savart law (17) in the stationary case. Equations (17)–(19) show that the conduction currents are the only primary source of the magnetic field. These are certainly cause–effect type laws.

A question closely related with the present issue is whether displacement current density in the space between the electrodes should be included in the *current* density in (17) and (19). One answer is that, if its effect is directly calculated by the Biot–Savart law, it turns out to be 0 [1]; see also [18] and the references therein. More fundamentally the displacement current density related with the conservative or irrotational electric field whose curl is zero such as that between the capacitor electrodes does not result in a magnetic field [6, 7, 10–12].

Bartlett [11] made an analytical calculation of the magnetic field between the capacitor plates to show with some approximation that it is actually created by the linear current in the lead wire and the radial current in the plates. Milsom [12] provided numerical results together with an excellent compact review of the topic. Previously the calculation had only been explained in principle [10].

Next, let us return to check the integral form of the Ampere–Maxwell law. The differential form of Maxwell's equations describe the relationship between the time and space derivatives of the electric and magnetic fields and the charge and current densities at a given position and time, and are valid at any place and time. It is possible to regard them as simultaneous

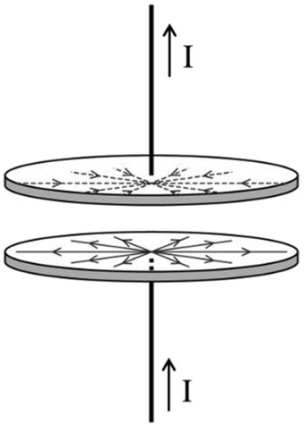


Figure 3. A parallel-plate capacitor being charged by a current I. Also shown are radial currents spreading out on the inside surface of the bottom plate, charging it positively, and those centering in on the inside surface of the top plate, charging it negatively.

differential equations, express them with a scalar and vector potentials, and get solutions (retarded potentials) which have forms showing causality. However, the integral form of the equations are not the integral solutions of the differential form. The former is derived by applying vector analysis to the respective latter with time variable t fixed throughout the whole space. For example, surface integration of both sides of the differential form of (6) on an arbitrary surface S bordered by any closed curve C yields

$$\int_{\mathrm{S}} (\nabla \times \boldsymbol{H}(\boldsymbol{r}, t)) \cdot \mathrm{d}\boldsymbol{S} = \int_{\mathrm{S}} \left(\boldsymbol{j}(\boldsymbol{r}, t) + \frac{\partial \boldsymbol{D}(\boldsymbol{r}, t)}{\partial t} \right) \cdot \mathrm{d}\boldsymbol{S}. \tag{21}$$

Then rewriting the left-hand side as a line integral along C from Stokes' theorem, we obtain the integral form of (6). That is, the integral form represents the correlation or association between the integral of the field $\boldsymbol{H}$ along C and that of $\boldsymbol{j} + \partial \boldsymbol{D}/\partial t$ on S *at a common fixed time* in an inertial frame of reference. Thus the integral form of the Ampere–Maxwell law cannot be regarded as a cause–effect type law. Moreover it is only when the conduction current and the displacement current distributions are highly symmetric that we may calculate the magnetic field at a point by reducing the integral along the path to a product such as $2\pi R H_1 = I$, leading to (10). It should be remembered, on the other hand, the law itself holds in the above sense even in a very rapidly changing time-dependent field regardless of the size of the integration domain. A subtle consequence is that the integral form of the Ampere–Maxwell law which represents the relationships of the fields at a given time could be applied to calculate approximately the magnetic field in a quasi-static case.

4. Reported measurements of displacement current

There have been many papers published whose titles include 'measurement of displacement current', 'effect of displacement current' and so on, as cited by Roche [2] and French [9], the oldest of which is probably by Thompson [21]. More recent articles include reference [22]. All these experiments, and likely many other reports on this topic, take it for granted that the

displacement current density, or time derivative of the electric field multiplied by ε_0, $\varepsilon_0 \boldsymbol{E}/\partial t$, in the space between the electrodes of a capacitor creates the magnetic field in and around it. Most of them measure in fact the electromotive force induced in toroidal search coils by the changing magnetic field in the space. Direct measurements of the magnetic field are also reported by Bartlett *et al* [23, 24]. It is apparent that none of the reports measure the displacement current itself directly. Only if it were independently established that the displacement current is really causing the magnetic field in these situations, could they be considered to be at best indirect measurements of the displacement current. As we discussed above, however, the magnetic field is actually created by the conduction current only and merely has a correlation with the displacement current created at the same time.

5. Two kinds of electric fields

If the displacement current density between the capacitor electrodes does not create a magnetic field, one might ask why the displacement current density in the Ampere–Maxwell law is essential for the existence of electromagnetic waves. To answer this question it is crucial to note that the electric field in Maxwell's equations contains two kinds of electric fields [25].

One is directly responsible for Gauss's law (4), or using (2)

$$\oint_{\mathrm{S}} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \frac{1}{\varepsilon_0} \int_{\mathrm{V}} \rho \, \mathrm{d}V, \qquad \nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0}. \tag{22}$$

We denote this electric field by $\boldsymbol{E}_{\mathrm{C}}$. As a matter of course it satisfies

$$\oint_{\mathrm{S}} \boldsymbol{E}_{\mathrm{C}} \cdot \mathrm{d}\boldsymbol{S} = \frac{1}{\varepsilon_0} \int_{\mathrm{V}} \rho \, \mathrm{d}V, \qquad \nabla \cdot \boldsymbol{E}_{\mathrm{C}} = \frac{\rho}{\varepsilon_0}. \tag{23}$$

This field is due to the electric charge, whose r^{-2} dependence around a point charge leads to its conservative or irrotational nature

$$\oint_{\mathrm{C}} \boldsymbol{E}_{\mathrm{C}} \cdot \mathrm{d}\boldsymbol{r} = 0, \qquad \nabla \times \boldsymbol{E}_{\mathrm{C}} = 0 \tag{24}$$

where, as before, C indicates arbitrary closed path in space.

The other electric field is directly responsible for Faraday's law (5). It is the electromagnetic induced field, and we denote it as $\boldsymbol{E}_{\mathrm{I}}$. It satisfies

$$\oint_{\mathrm{C}} \boldsymbol{E}_{\mathrm{I}} \cdot \mathrm{d}\boldsymbol{r} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathrm{S}} \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S}, \qquad \nabla \times \boldsymbol{E}_{\mathrm{I}} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{25}$$

where S, as before, is any surface bordered by any closed path C in space. The left-hand side of the integral form of (25), and also that of (5), represents the electromotive force $\mathcal{E}$ along one turn of C;

$$\varepsilon = \oint_{\mathrm{C}} \boldsymbol{E}_{\mathrm{I}} \cdot \mathrm{d}\boldsymbol{r} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathrm{S}} \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S}. \tag{26}$$

Since $\varepsilon \neq 0$, $\boldsymbol{E}_{\mathrm{I}}$ is not conservative, unlike $\boldsymbol{E}_{\mathrm{C}}$ which satisfies (24).

Maxwell's equations express the electric field simply by $\boldsymbol{E}$ without distinguishing $\boldsymbol{E}_{\mathrm{C}}$ and $\boldsymbol{E}_{\mathrm{I}}$. Since these fields are different and mutually independent as shown above, this means that

$$\boldsymbol{E} = \boldsymbol{E}_{\mathrm{C}} + \boldsymbol{E}_{\mathrm{I}}. \tag{27}$$

It is not possible to observe the two fields $\boldsymbol{E}_{\mathrm{C}}$ and $\boldsymbol{E}_{\mathrm{I}}$ separately by measurement since they have exactly the same property of electric field on a charge as defined by (1). But we are sure that the electric field in the absence of changing magnetic field is $\boldsymbol{E}_{\mathrm{C}}$ while that in the absence of electric charge is $\boldsymbol{E}_{\mathrm{I}}$.

In the following, we will examine the relationship between the electric field $\boldsymbol{E}$ appearing explicitly in Maxwell's equations and the two fields $\boldsymbol{E}_{\mathrm{C}}$ and $\boldsymbol{E}_{\mathrm{I}}$ that comprise it, and then point out that overlooking the particular property of $\boldsymbol{E}$ could lead to misunderstandings about the nature of the displacement current density addressed in this paper.

First we check Faraday's law of magnetic induction. Substituting (27) into (5) yields

$$\oint_{\mathrm{C}} (\boldsymbol{E}_{\mathrm{C}} + \boldsymbol{E}_{\mathrm{I}}) \cdot \mathrm{d}\boldsymbol{r} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathrm{S}} \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S}, \qquad \nabla \times (\boldsymbol{E}_{\mathrm{C}} + \boldsymbol{E}_{\mathrm{I}}) = -\frac{\partial \boldsymbol{B}}{\partial t}. \tag{28}$$

By subtracting (25) from this, we obtain

$$\oint_{\mathrm{C}} \boldsymbol{E}_{\mathrm{C}} \cdot \mathrm{d}\boldsymbol{r} = 0, \qquad \nabla \times \boldsymbol{E}_{\mathrm{C}} = 0 \tag{29}$$

which is the same as (24). The conservative or irrotational nature of the $\boldsymbol{E}_{\mathrm{C}}$, which is originally derived from the r^{-2} dependency of the field, is derived from (5), (25) and (27).

Because of (29) the law for $\boldsymbol{E}_{\mathrm{I}}$, (25), holds for E including $\boldsymbol{E}_{\mathrm{C}}$ as (5) without affecting the mathematical correctness of the equation. In other words, even though $\boldsymbol{E}_{\mathrm{C}}$ is included in the electric field $\boldsymbol{E}$ in equation (5), it is irrelevant to Faraday's law of magnetic induction.

Next let us check Gauss' law (4) with (2), i.e., (22). Substituting (27) into (22) yields

$$\oint_{\mathrm{S}} (\boldsymbol{E}_{\mathrm{C}} + \boldsymbol{E}_{\mathrm{I}}) \cdot \mathrm{d}\boldsymbol{S} = \frac{1}{\varepsilon_0} \int_{\mathrm{V}} \rho \, \mathrm{d}V, \qquad \nabla \cdot (\boldsymbol{E}_{\mathrm{C}} + \boldsymbol{E}_{\mathrm{I}}) = \frac{\rho}{\varepsilon_0}. \tag{30}$$

By subtracting (23) from this, we obtain

$$\oint_{\mathrm{S}} \boldsymbol{E}_{\mathrm{I}} \cdot \mathrm{d}\boldsymbol{S} = 0, \qquad \nabla \cdot \boldsymbol{E}_{\mathrm{I}} = 0. \tag{31}$$

This means that $\boldsymbol{E}_{\mathrm{I}}$ is divergence-free everywhere, unlike $\boldsymbol{E}_{\mathrm{C}}$ satisfying (24). In other words, even though $\boldsymbol{E}_{\mathrm{I}}$ is included in the electric field $\boldsymbol{E}$ in equation (4), it is irrelevant to Gauss' law. Note that equation (31) cannot be derived independently from the nature of $\boldsymbol{E}_{\mathrm{I}}$, but is deduced from within Maxwell's equations [25]. Helmholtz theorem guarantees that any vector field that goes to zero faster than $1/r$ as $r \to \infty$ can be decomposed into irrotational and divergence-free fields [26]. It now turns out that (27) represents such a decomposition of the electric field $\boldsymbol{E}$ in Maxwell's equations, which itself is neither irrotational nor divergence-free.

Next, let us check the Ampere–Maxwell law (7). Taking the divergence of both sides of its differential form after using (2) and (3), we get

$$\nabla \cdot (\nabla \times \boldsymbol{B}) = \mu_0 \nabla \cdot \boldsymbol{j} + \varepsilon_0 \mu_0 \frac{\partial}{\partial t} \nabla \cdot \boldsymbol{E}. \tag{32}$$

The left-hand side is zero, and substituting (27) for the second term on the right-hand side, we obtain

$$\nabla \cdot \boldsymbol{j} + \varepsilon_0 \frac{\partial}{\partial t} \nabla \cdot (\boldsymbol{E}_{\mathrm{C}} + \boldsymbol{E}_{\mathrm{I}}) = 0 \tag{33}$$

and taking (31) into account,

$$\nabla \cdot \boldsymbol{j} + \nabla \cdot \left(\varepsilon_0 \frac{\partial \boldsymbol{E}_\mathrm{C}}{\partial t} \right) = 0. \tag{34}$$

Referring to (23), it is evident that (34) reduces to the law of charge conservation (13). Equation (34) shows that $\varepsilon_0 \partial \boldsymbol{E}_\mathrm{C}/\partial t$ is the sole displacement current density effectively involved in (13). In other words, even though $\varepsilon_0 \partial \boldsymbol{E}_\mathrm{I}/\partial t$ is included in the displacement current density term $\partial \boldsymbol{D}/\partial t$ in (7), it is irrelevant to the law of charge conservation.

Finally let us consider the derivation of the differential equations for electromagnetic waves. If we take the curl of both sides of the differential form of (5), we have

$$\nabla \times \nabla \times \boldsymbol{E} = -\frac{\partial}{\partial t} \nabla \times \boldsymbol{B}. \tag{35}$$

Substituting (27) into the left-hand side of (35) yields

$$\nabla \times \nabla \times (\boldsymbol{E}_\mathrm{C} + \boldsymbol{E}_\mathrm{I}) = \nabla \times \nabla \times \boldsymbol{E}_\mathrm{I} = \nabla(\nabla \cdot \boldsymbol{E}_\mathrm{I}) - \nabla^2 \boldsymbol{E}_\mathrm{I} = -\nabla^2 \boldsymbol{E}_\mathrm{I}. \tag{36}$$

We have used first (24) and then the vector analysis identity and (31). From (7) with (2) and (3), the right-hand side of (35) becomes

$$-\frac{\partial}{\partial t} \nabla \times \boldsymbol{B} = -\mu_0 \frac{\partial \boldsymbol{j}}{\partial t} - \varepsilon_0 \mu_0 \frac{\partial}{\partial t} \left(\frac{\partial \boldsymbol{E}_\mathrm{C}}{\partial t} + \frac{\partial \boldsymbol{E}_\mathrm{I}}{\partial t} \right). \tag{37}$$

Thus we have

$$\nabla^2 \boldsymbol{E}_\mathrm{I} = \mu_0 \frac{\partial \boldsymbol{j}}{\partial t} + \varepsilon_0 \mu_0 \frac{\partial}{\partial t} \left(\frac{\partial \boldsymbol{E}_\mathrm{C}}{\partial t} + \frac{\partial \boldsymbol{E}_\mathrm{I}}{\partial t} \right). \tag{38}$$

Now remembering Helmholtz theorem, we decompose the current density into the sum of the irrotational (often called longitudinal) part $\boldsymbol{j}_\mathrm{L}$ and the divergence-free (often called transverse) part $\boldsymbol{j}_\mathrm{T}$

$$\boldsymbol{j} = \boldsymbol{j}_\mathrm{L} + \boldsymbol{j}_\mathrm{T} \tag{39}$$

where

$$\nabla \times \boldsymbol{j}_\mathrm{L} = 0 \quad \text{and} \quad \nabla \cdot \boldsymbol{j}_\mathrm{T} = 0. \tag{40}$$

Then (38) is separated into two:

$$\nabla^2 \boldsymbol{E}_\mathrm{I} - \varepsilon_0 \mu_0 \frac{\partial^2 \boldsymbol{E}_\mathrm{I}}{\partial t^2} = \mu_0 \frac{\partial \boldsymbol{j}_\mathrm{T}}{\partial t} \tag{41}$$

$$\frac{\partial}{\partial t} \left(\boldsymbol{j}_\mathrm{L} + \varepsilon_0 \frac{\partial \boldsymbol{E}_\mathrm{C}}{\partial t} \right) = 0. \tag{42}$$

Equation (41) is the inhomogeneous differential equation for the electric field part of the electromagnetic wave. It shows that $\varepsilon_0 \partial \boldsymbol{E}_\mathrm{I}/\partial t$ is the sole displacement current density effectively involved in the derivation of the electromagnetic waves. In other words, even though $\varepsilon_0 \partial \boldsymbol{E}_\mathrm{C}/\partial t$ which is related to the conservative electric field as that between the electrodes of a capacitor

is included in $\partial \boldsymbol{D}/\partial t$ in (7), it is irrelevant to the electromagnetic waves. Taking the divergence of (42) and using (23), we have

$$\frac{\partial}{\partial t}\left(\nabla \cdot \boldsymbol{j}_{\mathrm{L}} + \frac{\partial}{\partial t}\rho\right) = 0. \tag{43}$$

Noting that $\nabla \cdot \boldsymbol{j} = \nabla \cdot \boldsymbol{j}_{\mathrm{L}}$, this is the time derivative of the law of charge conservation (13), showing that (13) always holds.

For the magnetic field part of the wave equation, we take the curl of (7) with (2), (3), (27) and (39), obtaining

$$\begin{aligned}\nabla \times \nabla \times \boldsymbol{B} &= \mu_0 \nabla \times \left(\boldsymbol{j}_{\mathrm{L}} + \boldsymbol{j}_{\mathrm{T}}\right) + \varepsilon_0 \mu_0 \nabla \times \left(\frac{\partial \boldsymbol{E}_{\mathrm{C}}}{\partial t} + \frac{\partial \boldsymbol{E}_{\mathrm{I}}}{\partial t}\right) \\ &= \mu_0 \nabla \times \boldsymbol{j}_{\mathrm{T}} + \varepsilon_0 \mu_0 \nabla \times \frac{\partial \boldsymbol{E}_{\mathrm{I}}}{\partial t}. \end{aligned} \tag{44}$$

Taking into account (6) in the left-hand side after using the vector analysis identity as in (36), substituting (25) into the right-hand side, and then collecting the terms containing $\boldsymbol{B}$ on the left-hand side, we have the inhomogeneous wave equation for the magnetic field part of the electromagnetic wave

$$\nabla^2 \boldsymbol{B} - \varepsilon_0 \mu_0 \frac{\partial^2 \boldsymbol{B}}{\partial t^2} = -\mu_0 \nabla \times \boldsymbol{j}_{\mathrm{T}}. \tag{45}$$

Equation (44) shows, again, that $\varepsilon_0 \partial \boldsymbol{E}_{\mathrm{I}}/\partial t$ is the sole electric displacement current density effectively involved in the derivation of the electromagnetic waves.

6. Summary

Possible causes of the misconception about the nature of the displacement current density in Maxwell's equations are discussed. Firstly, an example of applying the integral form of the Ampere–Maxwell law to calculate the magnetic field in and around a parallel-plate capacitor by using a plane between the electrodes and parallel to the electrodes may lead to the misunderstanding that the displacement current density between the capacitor electrodes could be a source of the magnetic field. It is urged that students' attention is drawn to the fact that the choice of surface used in the calculation does not, of course, determine the source of the magnetic field obtained. Secondly, misinterpretation of the integral form of the Ampere–Maxwell law as expressing a cause–effect relationship could cause confusion. It is in fact a formula showing a correlation or association of the fields. Thirdly, it has been pointed out that the electric field $\boldsymbol{E}$ appearing in Maxwell's equations is the sum of the conservative field $\boldsymbol{E}_{\mathrm{C}}$ and the induction field $\boldsymbol{E}_{\mathrm{I}}$. How they are involved in Maxwell's equations has been carefully explored with the aim of identifying the underlying cause of the controversies concerning the displacement current. It is shown, in particular, that only the displacement current density related with $\boldsymbol{E}_{\mathrm{C}}$ makes Maxwell's equations consistent with the law of charge conservation, while only that related with $\boldsymbol{E}_{\mathrm{I}}$ plays a crucial role in the derivation of the differential equations for electromagnetic waves. Furthermore, additional support provided from the calculations using the Biot–Savart law which show that the magnetic field between the capacitor plate is actually created by the real currents alone have only recently been reported. This late confirmation may have been another factor which allowed the misconception to persist for a long time.

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Topic: Maxwell at the nanoscale

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Parity-time-symmetric plasmonic metamaterials

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PT-Symmetric Plasmonic Metamaterials

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We theoretically investigate the optical properties of parity-time (PT) symmetric three-dimensional metamaterials composed of strongly-coupled planar plasmonic waveguides. By tuning the loss-gain balance, we show how the initially isotropic material becomes both asymmetric and unidirectional. Investigation of the band structure near the material's exceptional point reveals several intriguing optical properties, including double negative refraction, Bloch power oscillations, unidirectional invisibility, and reflection and transmission coefficients that are simultaneously equal to or greater than unity. The highly tunable optical dispersion of PT-symmetric metamaterials provides a foundation for designing an entirely new class of three-dimensional bulk synthetic media, with applications ranging from lossless sub-diffraction-limited optical lenses to non-reciprocal nanophotonic devices.

Textbook conceptions of light-matter interactions have been challenged by two recent material advances — the development of metamaterials and the discovery of parity-time PT-symmetric media. Metamaterials allow considerable control over the electric and magnetic fields of light, so that permittivities, permeabilities, and refractive indices can be tuned throughout positive, negative, and near-zero values. Metamaterials have enabled negative refraction, optical lensing below the diffraction limit of light and invisibility cloaking [1–9]. Complementarily, PT-symmetric media allow control over electromagnetic field distributions in loss and gain media, so that light propagation can be asymmetric and even unidirectional. PT-symmetric media have enabled loss-induced optical transparency, lossless Talbot revivals and unidirectional invisibility [10–21]. Combined with non-linear media, they have also been suggested as optical diodes, insulators, circulators, and perfect cavity absorber-lasers [22–28].

While metamaterials rely on subwavelength engineered 'building blocks' to control electric and magnetic light-matter interactions, PT-symmetric media rely on judicious spatial arrangement of loss and gain media. Their unique asymmetric properties are based on a fundamental insight from quantum mechanics indicating that Hamiltonians need not be Hermitian to yield real eigenvalues and hence physical observables. Instead, the weaker condition of parity and time symmetry is sufficient to yield real eigenvalues below a certain threshold. Above this threshold, eigenvalues move into the complex plane and become complex conjugates of each other [29–33].

In the context of optics, PT-symmetric Hamiltonians arise from the duality between the quantum mechanical Schrodinger equation and the wave equation. Provided the refractive index profile satisfies $n(x) = n*(-x)$, light will propagate as if it experiences a PT-symmetric potential. Below the PT-symmetric exceptional point, the optical eigenvalues will be purely real; however, as the loss and gain of the material are increased beyond the exceptional point, the eigenvalues will become complex. In particular, certain eigenmodes will experience increased loss while other eigenmodes will exhibit strong optical gain. This behavior is at the core of the asymmetric and unidirectional optical properties observed in PT media to date.

While nearly all PT-symmetric media have been constructed from macroscopic (i.e., greater than wavelength-scale) elements, the optical Hamiltonian places no restrictions on the length scales over which the index profile can vary. This insight drives the question: can we create PT-symmetric metamaterials — i.e., bulk photonic media whose optical properties are determined both by their subwavelength building blocks and a judicious choice of their loss/gain profile? Such metamaterials would enable unprecedented control over electric and magnetic optical fields across wavelength and subwavelength scales, and may enable an entirely new class of bulk synthetic photonic media.

In this Letter, we investigate the emergent optical properties of bulk, three-dimensional PT-symmetric metamaterials. As a prototype metamaterial, we consider a multilayer stack of alternating layers of metal and dielectric. Both theoretical [34] and experimental [6] work has demonstrated the isotropic negative index response of this metamaterial, resulting in all-angle negative refraction and Veselago 'perfect optical lensing.' Its operation is based on the negative index plasmonic modes of its unit cell — a five-layer 'metal-insulator-metal' waveguide. By varying the thickness of the layers as well as the materials, the frequency of operation and the emergent bulk index of refraction can be precisely controlled throughout optical frequencies. While practical utilization of this negative index metamaterial has been limited by propagation and coupling losses, we will show that these losses could be overcome by subjecting the plasmonic modes to PT-symmetric potentials. Moreover, PT-symmetric potentials in this metamate-

rial can enable above-unity transmission and reflection, Bloch power oscillations, hyperbolic to elliptic dispersion transitions, and unidirectional invisibility.

Figure 1a illustrates the specific plasmonic metamaterial investigated in this paper, with the unit cell period indicated by Λ. Within each unit cell, the thicknesses of the metal t_m and dielectric t_d are deeply subwavelength, with $t_m = t_d = 30$ nm. We consider Ag as the metal, described by a lossless Drude model with dielectric constant $\epsilon_{Ag} = 1-(\frac{\omega_p}{\omega})^2$ [35] The bulk plasma frequency of Ag, ω_p, is assumed to be $8.85 \times 10^{15}\text{s}^{-1}$. We consider the dielectric layers to be TiO_2 with n = 3.2. With these materials, the surface plasmon resonance, ω_{sp}, occurs at 1.73 eV ($\omega_{sp}/\omega_p = 0.29$), and negative index modes are observed between this frequency and ω_p. This particular materials combination was recently experimentally shown to exhibit all-angle negative refraction and Veselago lensing [6]. Here, we theoretically investigate the evolution of the optical bands of this metamaterial upon varying the imaginary part of the refractive index of TiO_2, denoted as k in Fig.1. PT-symmetric potentials require balanced loss and gain, so the magnitude of this 'non-Hermiticity parameter', k, is identical for alternating dielectric layers.

Using the transfer matrix approach described in [36], we solve for the dispersion curves of the five-layer unit-cell plasmonic waveguide for transverse-magnetic (TM) polarized illumination. To determine the band diagrams of the periodic metamaterial, the wavevector along the z-direction is swept in the first Brillouin zone, $(0, \frac{\pi}{\Lambda})$, and the characteristic equation is minimized to find the propagation constant along the x-direction at each frequency. The results are shown in panels (b)-(e) of Fig. 1 for k = 0, 0.2, 0.3 and 0.5, respectively. Note that the colormap indicates purely real values of k_x, corresponding to lossless propagation along the metamaterial. For a non-Hermiticity parameter $k = 0$, four different branches are observed: two below ω_{sp} (B_1 and B_2) and two above (B_3 and B_4). Because all constituents are lossless, the wavevectors diverge at ω_{sp}. B_1 and B_2 are characterized by positive slopes and hence positive refractive mode indices. In contrast, B_3 and B_4 are characterized by negative slopes and hence negative refractive mode indices.

When the non-Hermiticity parameter of the metamaterial is increased, the modes merge together at the exceptional points of the dispersion, denoted by black circles in panels (c)-(e). Beyond these exceptional points, the two distinguishable lossless modes below and above ω_{sp} (i.e., B_1 and B_2 or B_3 and B_4, respectively) evolve to a gain mode and a loss mode with the same phase velocity. Due to their complex wavevectors, we denote these modes as black, dashed lines in Fig. 1(c)-(e). To understand these loss and gain modes, note that the transfer matrix of the PT-symmetric metamaterial possesses the following symmetry property:

$$T(\omega, k_z, k_x^*)T^*(\omega, k_z, k_x) = I \qquad (1)$$

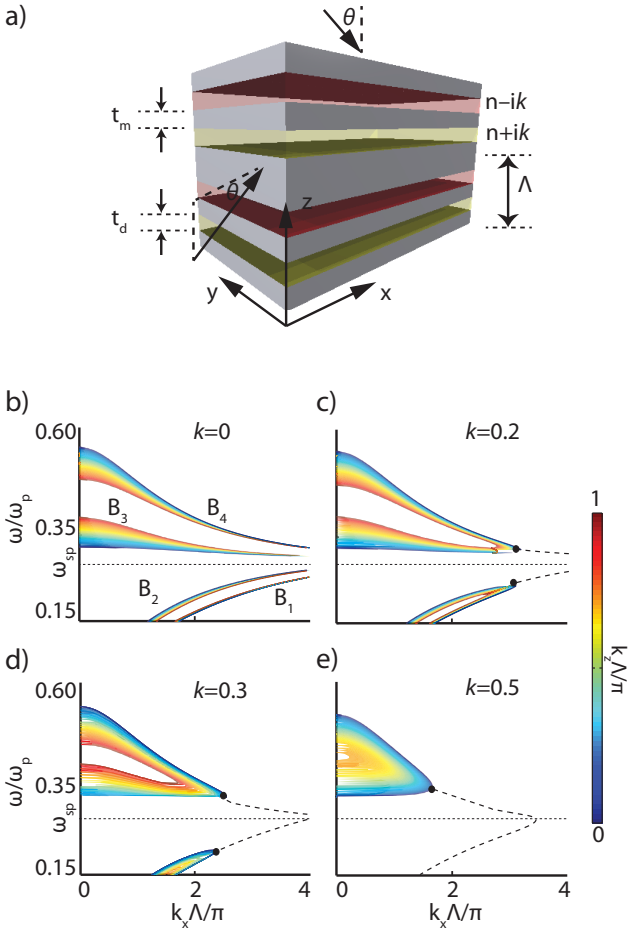


FIG. 1. Schematic of the plasmonic metamaterial (a), and the associated band diagrams for transverse magnetic modes (TM). The metamaterial is composed of coupled plasmonic waveguides, with deeply subwavelength metal and dielectric thicknesses of t_m=t_d= 30 nm. The unit-cell period is Λ = 150 nm. The refractive index of the dielectric is $n \pm ik$, with $n = 3.2$ and k corresponding to the non-Hermiticity parameter. Band diagrams are included for k= b)0, c)0.2, d)0.3 and e)0.5. Bands B_1 and B_2 are positive refractive index bands, while bands B_3 and B_4 correspond to negative refractive index bands. The horizontal dotted black lines in panels (b)-(e) indicate the surface plasmon resonance frequency, ω_{sp}. The black circles in panels in the same panels indicate the exceptional points of the dispersion, beyond which the purely real modes evolve to the complex modes characterized by either loss or gain (dashed black lines).

where I is the identity matrix. The Bloch modes of the metamaterial are eigenvalues of T and satisfy:

$$|T(k_x) - e^{i\Lambda k_z} I| = 0 \qquad (2)$$

Taking the complex conjugate of Eq. 2 and using the symmetry property of Eq. 1, the following relation is ob-

tained:

$$|T(k_x^*) - e^{i\Lambda k_z} I| = 0 \tag{3}$$

Equation 3 means that if k_x (a complex number in general) admits a real solution for the Bloch wavevector, k_x^* is a solution for that Bloch mode as well. Accordingly, the bands have centro-symmetry in the complex (k_x, k_z) plane. Also note that the loss and gain modes of Fig. 1 (c)-(e) conjoin at ω_{sp}; further, unlike the modes for a zero non-Hermiticity parameter, their wavevectors at ω_{sp} remain finite.

While *real* periodic spatial refractive index profiles lead to the appearance of an infinite number of band gaps, *complex* periodic index profiles generally result in complex dispersion curves across the entire frequency range. Interestingly, if the refractive index profile satisfies the condition for PT symmetry ($n(z) = n^*(-z)$) real propagation constants and complete band gaps can exist provided $k \leq k_{th}$. Here, k_{th} is the threshold value at which the Hamiltonian and the PT operator no longer commute, and consequently, real-valued solutions cease to be supported by the complex potential. Fig. 1 (c)-(e) illustrate this feature for increasing non-Hermiticity parameter. For example, for $k = 0.2$ and $k = 0.3$ purely real wavevectors and bandgaps are observed for all bands both above and below ω_{sp}. However, for $k = 0.5$, purely real eigenmodes below ω_{sp} do not exist across visible and near-infrared frequencies. Further, the bandgap between B_3 and B_4 merges for large k_z, and these bands only exist over a very limited wavevector and wavelength range.

The non-Hermiticity parameter not only changes the propagation constant and bandgap of the metamaterial, but also the band curvature. Fig. 2 plots the equi-frequency contours of bands B_1-B_4 at wavelengths of $\lambda = 954$ nm ($\omega/\omega_p = 0.22$) for B_1 and B_2, $\lambda = 604$ nm ($\omega/\omega_p = 0.35$) for B_3, and $\lambda = 445$ nm ($\omega/\omega_p = 0.48$) for B_4. Since the metamaterial is isotropic in the xy-plane and the contours are centro-symmetric in the (k_z, k_x) plane, a quadratic dispersion relation $(\frac{k_x}{n_x})^2 + (\frac{k_z}{n_z})^2 = k_0^2$ can be used to model the bands. Here k_0 indicates the free-space wavevector. The fitted refractive mode indices are listed in Table I.

	$k = 0$	$k = 0.2$	$k = 0.5$
	$(n_x^2\ ,\ n_z^2)$	$(n_x^2\ ,\ n_z^2)$	$(n_x^2\ ,\ n_z^2)$
B_1	(95.86 , 289.89)	(85.5 , 53.56)	NA
B_2	(54.14 , −58.86)	(59.32 , −24.83)	NA
B_3	(−4.11 , 1.27)	(−4.88 , 1.21)	(11.55 , 1.16)
B_4	(1.9 , 1.85)	(1.85 , 1.45)	(1.02 , 0.66)

TABLE I. Effective refractive indices of the four bands based on a quadratic fit. The parameters are calculated for wavelengths of $\lambda = 954$ nm (B_1,B_2), 604 nm (B_3) and 445 nm (B_4).

As seen both in Fig. 2 and Table I, for a non-Hermiticity parameter $k = 0$, bands B_1 and B_4 are elliptical (i.e., $n_x^2 n_z^2 \geq 0$) while bands B_2 and B_3 are hyperbolic (i.e., $n_x^2 n_z^2 \leq 0$). Moreover, B_4 is characterized by a nearly-perfect circular equi-frequency contour and almost equal values of effective refractive indices in both the x and z-directions. Accordingly, this metal-insulator-metal metamaterial is isotropic at $\lambda = 445$ nm, consistent with prior work [6, 34].

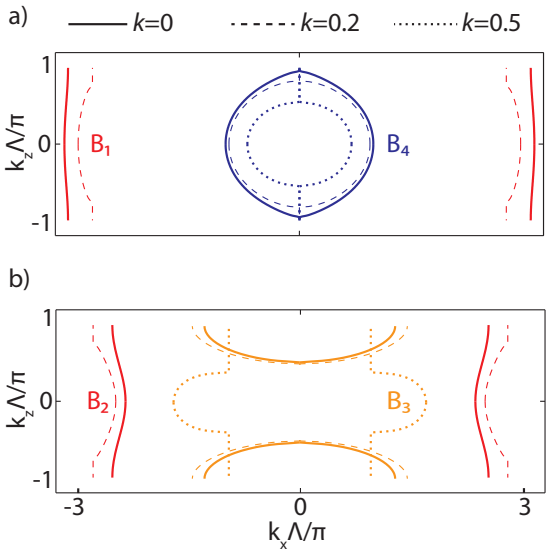


FIG. 2. Equi-frequency contours of the metamaterial, plotted for both positive and negative index bands at wavelengths of $\lambda = 954$ nm (B_1,B_2), 604 nm (B_3) and 445 nm (B_4). Band B_4 is circular for zero and small k, corresponding to an isotropic metamaterial. For all k, bands B_1 and B_4 remain elliptical while band B_2 is hyperbolic. However, band B_3 undergoes a hyperbolic to elliptical transition with increasing k.

For increasing non-Hermiticity parameter, B_1 and B_4 remain elliptical while B_2 remains hyperbolic. Interestingly, however, band B_3 undergoes a hyperbolic to elliptical transition for $k = 0.5$. Such hyperbolic-to-elliptic transitions could enable dynamic tuning of Purcell enhancements for emitters near the metamaterial. Further, they could modulate Talbot revivals or the formation and resolution of images generated by hyperbolic metamaterial super-lenses [37–41].

The results of Fig. 2 imply that with increasing non-Hermiticity parameter, the material can evolve from an isotropic metamaterial to an anisotropic one. Intriguingly, the structure can also become highly directional. This property cannot be derived from the band diagrams, but can be understood by considering the transfer matrix:

$$T = \begin{bmatrix} a & b \\ c & a^* \end{bmatrix} \tag{4}$$

Here, the parameters a, b and c are related to the reflec-

tion and transmission coefficients r and t as $r_L = -\frac{c}{a^*}$, $r_R = \frac{b}{a^*}$ and $t_L = t_R = \frac{1}{a^*}$, where the subscripts L and R denote illumination from the left and right, respectively. As these equations indicate, an optical system composed of linear and reciprocal materials is non-directional provided the components are lossless. In other words, the transmitted and reflected powers $T = |t|^2$ and $R = |r|^2$ sum to unity and are independent of illumination direction, since $T_L = T_R = T$ and $T + R_R = 1 = T + R_L$, so $R_L = R_R$. When loss or gain is introduced into the system, the transmission coefficient remains the same for both directions of illumination. However the reflection coefficient need not be symmetric, as power can be attenuated or generated within the structure. The asymmetry is obtained at the price of losing propagating Bloch modes. However, as we will show, asymmetric responses can be obtained in a PT-symmetric potential where purely real bands exist as well.

To illustrate this directional behavior, Fig. 3 plots plane-wave refraction of light from vacuum ($n = 1$) through a metamaterial composed of 10 unit cells. We consider TM-polarized illumination of wavelength $\lambda = 445$ nm impinging on the metamaterial at an angle of $\theta = 45°$ in the (x, z) plane. The colormap of Fig. 3 plots the H_y component of the fields. The arrows of Fig. 3 indicate the direction of illumination, refraction and transmission, each determined by spatially-averaging the Poynting vector in each region.

For $k = 0$ (Fig. 3a), the power is negatively refracted with an angle of $\sim -32°$. This result is in excellent agreement with our band structure calculations, which yield a refracted angle from Snell's law of $\sim -31°$. The refractive index, $n = -\sqrt{1.87} = -1.36$, at this non-Hermiticity value is independent of the illumination angle and direction. Indeed, for illumination in the (x, z) plane, or an 'endfire configuration', the same refraction angle is observed (see Fig. 4a). The same refraction angle is also observed for illumination from all sides of the metamaterial (i.e., illumination from $\pm x$, $\pm y$, and $\pm z$)

Upon increasing the non-Hermiticity parameter of the metamaterial, the material becomes highly directional. Panels (b) and(c) of Fig. 3 illustrate the field profiles in a 10-layer PT-symmetric metamaterial when illumination is from the loss and gain side (i.e., illumination in the $+z$ or $-z$ directions, respectively). As a particular example, we consider $k = 0.445$. As seen, field profiles and refraction angles are completely different for illumination from $+z$ (loss side) versus $-z$ (gain side). Illumination from $+z$ yields negative refraction at an angle of $-81°$ (see Fig. 3b). In contrast, illumination from $-z$ yields negative refraction at an angle of $\sim -43°$ (Fig. 3c).

More intriguingly, this structure is characterized by tunable reflection and transmission coefficients that can range from zero to at or above unity. This characteristic is illustrated in the lower panels of Fig. 3, which plot the normalized-to-incidence power at each position along the

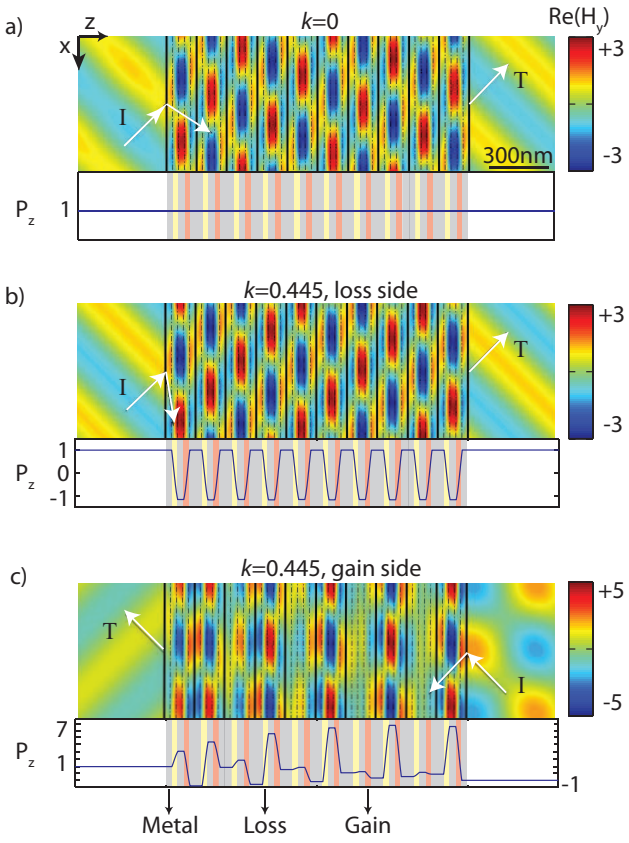


FIG. 3. Real(H_y) for a 10-unit-cell metamaterial upon illumination by a planewave in air of wavelength $\lambda = 445$ nm and angle $\theta = 45°$. Panel (a) shows the results for $k = 0$ while (b) and (c) show the results for $k = 0.445$ when illuminated from loss ($+z$) and gain ($-z$) side, respectively. The overlaid arrows indicate the direction of the averaged Poynting vectors in the corresponding regions. I in each panel indicates the *incident* wavevector while T shows the *transmitted* planewave. The lower graphs in each panel show the distribution of the power in each layer of the stack. The metallic layer is represented in gray, while the gain and loss regions are represented in red and yellow, respectively.

direction of propagation. For example, for illumination from the $-z$ direction (panel (c)) power flows backward toward the source in the illumination region ($P_z = -1$), and away from the metamaterial on the transmission side ($P_z = +1$). Therefore, the metamaterial is completely transparent, in that the metamaterial can transmit all of the incident power, even though light is emitted back towards the source.

Moreover, for illumination from the $+z$ direction (panel (b)) this metamaterial can achieve unidirectional invisibility. As seen, the power is unity on both sides of the metmaterial, indicating complete suppression of reflection on the illumination side and complete trans-

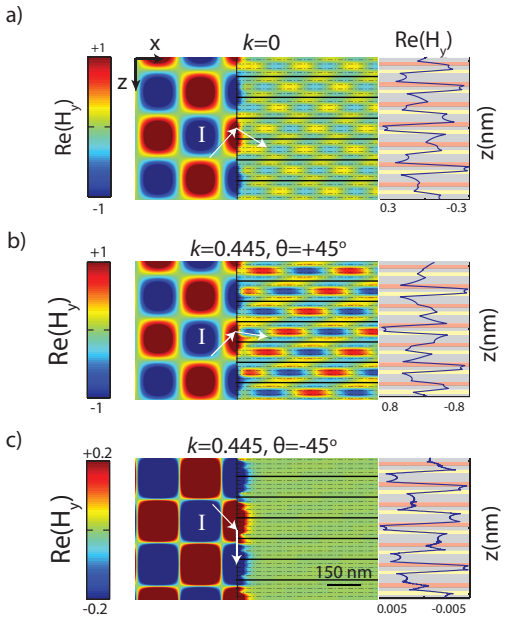


FIG. 4. Real(H_y) when a semi-infinite metamaterial is illuminated with a planewave at $\lambda = 445$ nm along x-direction (endfire illumination). The illumination angle θ equals 45° in (a) and (b) and −45° in (c). The side graphs in each panel show the distribution of the magnetic field 1μm away from the interface.

mission on the other. Formally, perfect invisibility requires that the transmitted phase (ϕ_t) equal the phase of a planewave propagating in free-space (ϕ_{FS}). For the 10-layer metamaterial of Fig. 3b, $\frac{\phi_{FS}-\phi_t}{2\pi} = 2.75$, so the object could be identified through the interference with a reference planewave. However, perfect unidirectional invisibility, i.e. $\phi_{FS} - \phi_t = 2m\pi$ where m is an integer, can be achieved when the number of periods is increased to 55, 74 and 91.

The unusual directional scattering properties of the metamaterial can be rationalized by considering the power as light propagates through the array. For $k = 0$, power remains constant throughout the metamaterial (Fig.3a). However, with increasing k, power begins to oscillate within the metamaterial, with power increasing in the gain regions and decreasing in the loss regions. These plasmonic power oscillations are analogous to Bloch oscillations observed in both electronic and photonic crystals [42] as well as PT-symmetric arrays [22].

As a final visual example of the unusual unidirectional properties of this metamaterial, Fig. 4 plots the fields and refracted angles for illumination along the $+x$ direction (endfire illumination). For a non-Hermiticity parameter $k = 0$, illumination at $\theta = \pm 45°$ yields refraction at $\mp 30°$, respectively, in good agreement with the previously determined value of $\mp 31°$. With increasing non-Hermiticity parameter, however, illumination at $+\theta$ yields markedly different results than illumination at $-\theta$. For example, for $k = 0.445$, illumination at +45° yields a refracted angle of $-11°$, while illumination at −45° yields refraction along the metamaterial interface at an angle of −90°. This double refraction does not just manifest itself in the intensity of the transmitted beam, but also in the profile of the fields, as seen in the right panels of Fig. 4.

We now consider the effect of varying the non-Hermiticity parameter on the scattering properties of the metamaterial. As before, we consider TM-polarized illumination with a 45° tilted planewave at $\lambda = 445$ nm. We limit our analysis to illumination along either $+z$ ('left-side illumination (L)') or $-z$ ('right-side illumination (R)') Based on Eq. 1, a generalized energy conservation formula can be derived as $|T-1| = \sqrt{R_R R_L}$, where T is the transmitted power and R_R and R_L are the reflected powers from the right and left sides, respectively [43].

Figure 5 plots the reflection and transmission coefficients as a function of k. As seen in Fig. 5d, for $k = 0.035$, the transmitted power equals unity independent of the number of layers. Correspondingly at this point R_R, shown in panel (b), vanishes for any number of layers. This property manifests itself as a peak in Fig. 4c, where the quotient of relative reflection coefficients is plotted. Importantly, for $k = 0.035$, this PT-symmetric metamaterial is still isotropic, characterized by circular equifrequency contours. Therefore, this PT-symmetric optical potential could enable lossless and far-field Vesalago lensing.

For larger non-Hermiticity parameters ($k >0.035$), T exceeds unity. Interestingly, while the transmitted power varies with k and the number of unit cells, it never drops below 1 up through $k = 0.445$. At this non-Hermiticity value, the reflected power from the left/loss side (R_L) vanishes, as shown in Fig. 5a. For larger k, T remains at or below unity. Non-Hermiticity parameters k above 0.63 yield purely imaginary k_z, so no propagation is allowed through the metamaterial. This property is accompanied by a rapid drop in T and strong increase in the reflectance for any number of layer.

While for left- (or loss)-side illumination the reflection coefficient is always less than unity, for right- (or gain)-side illumination the reflected power can exceed 1. This behavior is not monotonic however, depending strongly on the number of unit-cell layers and k. Indeed, only at $k = 0$ and $k = 0.12$ are the reflected powers from the left and right identical and equal to unity. For $k >0.12$, the ratio r_L/r_R remains below unity (Fig. 5c).

Table II summarizes some of the interesting scattering properties of the 10-unit-cell metamaterial extracted from Fig. 5. For $k = 0$, the metamaterial is Hermitian and lossless. With increasing PT-symmetric potentials, unusual features such as above unity transmission ($k = 0.26$), above unity reflection ($k = 0.445$) and uni-

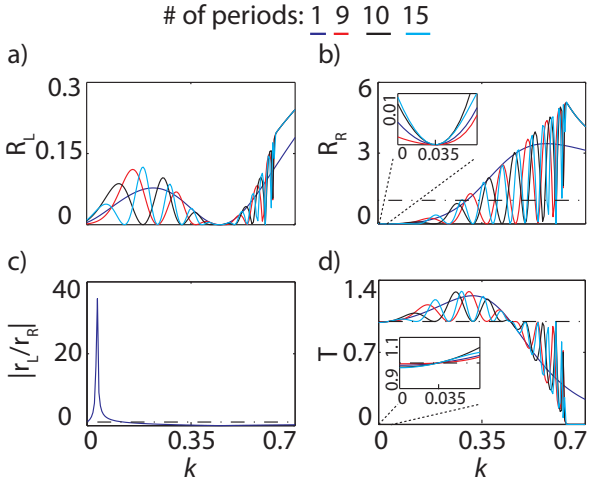


FIG. 5. Reflected and transmitted powers of the metamaterial with increasing k and varying numbers of periods. (a) Reflected power upon illumination from the left (loss) side of the metamaterial; (b) reflected power upon illumination from the gain (right) right side; (c) Quotient of reflected powers; (d) Transmitted power. In all panels, the incident angle is $\theta = 45^\circ$ and the illumination wavelength is $\lambda = 445$ nm. The inset in panel (b) and (d) show the behavior of the reflected and transmitted powers around $k = 0.035$ where R_R vanished and $T = 1$ independent of the number of periods.

directional reflection suppression ($k = 0.445$ and $k = 0.5$) are observed. The normalized Bloch wavevector for each k is included in the last row of the table. Since k_z is purely real, light propagates in a pass band of the metamaterial for all of these non-Hermiticity parameters.

	$k=0$	$k=0.035$	$k=0.26$	$k=0.445$	$k=0.5$
R_L	0.0080	0.03	0.0982	0	5×10^{-4}
R_R	0.0080	0	0.8397	2	0.1219
T	0.9920	1	1.2872	1	0.9922
$\frac{\Delta k_z}{\pi}$	0.7633	0.7609	0.6502	0.4683	0.3948

TABLE II. Transmitted and reflected powers from left and right and the normalized Bloch wavevector for 10-unit-cell metamaterial at $\lambda = 445$ nm and $\theta = 45^\circ$.

In conclusion, we have introduced the concept of a PT-symmetric metamaterial. The original lossless metamaterial, composed of a periodically-stacked 5-layer plasmonic waveguide, was designed to behave as an isotropic, three-dimensional negative refractive index material. By subjecting the plasmonic modes to PT-symmetric optical potentials, we demonstrated the broad tunability of the band curvatures, band gaps and effective refractive indices of the material. Small but non-zero non-Hermiticity parameters increased the transmission of the isotropic negative index metamaterial to unity. Larger non-Hermiticity parameters morphed the material from isotropic to anisotropic and directional. The highly unusual optical properties of PT-symmetric metamaterials could be used to devise an entirely new class of bulk synthetic media, ranging from lossless Veselago lenses to unidirectional metamaterial-based invisibility cloaks and new non-reciprocal nanophotonic devices.

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